



- ① 题: 可能仅 20% 证明
 ② 经常翻阅 yf I, 记结论

PI 微分学 (40%)

1. 二重极限

$P \rightarrow P_0$ 的任意性

$\Rightarrow$ 等价无穷小等方法

eg 1: $\lim_{x,y \rightarrow (0,0)} (x^2+y^2)^{\arcsin(x^2y)}$

设为 t . $\ln t = \lim_{x,y \rightarrow (0,0)} \arcsin(x^2y) \ln(x^2+y^2)$

尽可能使用极坐标: $= \lim_{(x,y) \rightarrow (0,0)} x^2y \ln(x^2+y^2) < \text{等价无穷小}>$
 $= \lim_{x,y \rightarrow (0,0)} \rho^3 \sin^2\theta \cos\theta \cdot 2 \ln \rho = 0$ (由 $\lim_{x \rightarrow 0} x^\alpha \ln x = 0, \alpha > 0$ 得)

则 $t = 1$

eg 2: $\lim_{(x,y) \rightarrow (\infty, \infty)} \frac{e^{-(\frac{y}{x} + \frac{x}{y})}}{(\frac{y}{x} + \frac{x}{y})^{xy}}$

显然 $\frac{x}{y} + \frac{y}{x} \geq 2$.

则 $0 \leq f(x,y) \leq \frac{e^0}{2^{xy}} = \frac{1}{2^{xy}}$

由夹逼定理, $\lim_{x,y \rightarrow (\infty, \infty)} f(x,y) = 0$

eg 3. (讨论在间断点连续性, 可推导性, 可微性, $\Rightarrow$ 也要用二重极限)

讨论 $f(x,y) = \begin{cases} (x-y) \arctan(\frac{1}{x^2+y^2}) & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$

讨论之. $(0,0)$ 处

$\begin{cases} \text{连续, 可微: 二重极限} \\ \text{可偏导} \end{cases}$

连续性: $\lim_{x,y \rightarrow (0,0)} (x-y) \arctan(\frac{1}{x^2+y^2}) = 0 = f(0,0)$ < 无穷小与有界变量之积 >

则 $f(x,y)$ 在 $(0,0)$ 连续.

可偏导性: $\frac{\partial f}{\partial x} = \lim_{x \rightarrow 0} \frac{f(x,0) - f(0,0)}{x} = \lim_{x \rightarrow 0} \arctan(\frac{1}{x}) = \frac{\pi}{2}$

$\frac{\partial f}{\partial y} = \lim_{y \rightarrow 0} \frac{f(0,y) - f(0,0)}{y} = \lim_{y \rightarrow 0} -\arctan(\frac{1}{y}) = -\frac{\pi}{2}$. 可偏导.

$$w = f(x, y) - f(0, 0) - f'_x \cdot x - f'_y \cdot y = (x-y) \arctan(\frac{1}{x+y}) - \frac{\pi}{2}x + \frac{\pi}{2}y$$

$$= (x-y) \left[\arctan(\frac{1}{x+y}) - \frac{\pi}{2} \right]$$

若 $\lim_{x \rightarrow 0, y \rightarrow 0} \frac{w}{\rho} = 0$, 则可微

$$\frac{w}{\rho} = \frac{\rho(\sin\theta - \cos\theta) \left[\arctan(\frac{1}{\rho}) - \frac{\pi}{2} \right]}{\rho} = (\sin\theta - \cos\theta) \cdot 0 = 0, \text{ 因此可微}$$

2. 偏导数 (复合函数 & 隐函数)

eg1: $y = y(x), z = z(x)$. 由方程 $\begin{cases} x^3 + y^3 + z^3 - 3xyz = 0 \\ x + y + z = a. \end{cases}$ 确定.

$$\text{求 } \frac{\partial y}{\partial x}, \frac{\partial z}{\partial x}.$$

$$\text{两边对 } x \text{ 求导: } \begin{cases} 3x^2 + 3y^2 y' + 3z^2 z' - 3(yz + xzy' + xy'z) = 0 \\ 1 + y' + z' = 0 \end{cases}$$

$$\text{解方程得 } \begin{cases} y' = \frac{z-x}{y-z} \\ z' = \frac{x-y}{y-z} \end{cases}$$

eg2: $f(u, v)$ 有 $f''_{uu} + f''_{vv} = 1$

$$g(x, y) = f(xy, \frac{1}{2}(x^2 - y^2)), \text{ 求 } g''_{xx} + g''_{yy}$$

$$\text{记 } u = xy, v = \frac{1}{2}(x^2 - y^2)$$

$$g'_x = \frac{\partial f}{\partial u} \cdot \frac{du}{dx} + \frac{\partial f}{\partial v} \cdot \frac{dv}{dx} = y \cdot \frac{\partial f}{\partial u} + x \cdot \frac{\partial f}{\partial v}$$

$$g'_y = \frac{\partial f}{\partial u} \cdot x + (-y) \cdot \frac{\partial f}{\partial v}$$

$$g''_{xx} = y \cdot [f''_{uu} \cdot y + f''_{uv} \cdot x] + x \cdot [f''_{vu} \cdot y + f''_{vv} \cdot x]$$

$$g''_{yy} = x [f''_{uu} \cdot x + f''_{uv} \cdot (-y)] + (-y) \cdot [f''_{vu} \cdot x + f''_{vv} \cdot (-y)]$$

$$\text{则 } g''_{xx} + g''_{yy} = f''_{uu}(x^2 + y^2) + f''_{vv}(x^2 + y^2) = x^2 + y^2.$$

3. 偏导几何应用

★一定要求切法向量

eg1: 求 $x^2 + 2y^2 + z^2 = 1$ 上平行于平面 $x - y + 2z = 0$ 的切平面.

设切于 (x_0, y_0, z_0) , 由题平面 写为 $(x - x_0) - (y - y_0) + 2(z - z_0) = 0$
 则法向量 $\vec{n} = (x_0, 2y_0, z_0)$, 平面法向量 $\vec{n}_1 = (1, -1, 2)$ $x_0x + 2y_0y + z_0z = 1$

$$\text{则: } \begin{cases} \vec{n} \parallel \vec{n}_1 \Rightarrow \text{可设 } \begin{cases} x = \frac{2}{\lambda} \\ y = -\frac{1}{\lambda} \\ z = \lambda \end{cases} \\ x_0^2 + 2y_0^2 + z_0^2 = 1 \end{cases}$$

$$\text{得 } \lambda = \pm \sqrt{\frac{8}{11}}, \text{ 化简: } x - y + 2z = \pm \sqrt{\frac{11}{2}}$$

4. 多元函数极值: $\begin{cases} 1. \text{无条件} \\ 2. \text{有条件} \end{cases}$

1. 求定义域

2. 求偏导得可疑点

3. 求三个二阶偏导, 计算 $\begin{cases} A = f''_{xx} \\ B = f''_{xy} \\ C = f''_{yy} \end{cases}$

$$B^2 - AC \begin{cases} > 0 & (\times) \\ = 0 & (\text{不可}) \\ < 0 & \Rightarrow \text{看 } A \text{ 或 } C \end{cases}$$

$\begin{cases} < 0 & \text{极大} \\ > 0 & \text{极小} \end{cases}$

($\neq 0$ 就是极值)

eg1: 求 $f(x, y) = (1 + e^y) \cos x - y e^y$ 之极值

$$\begin{cases} z'_x = -(1 + e^y) \sin x \\ z'_y = \cos x e^y - (y + 1) e^y = e^y (\cos x - y - 1) \end{cases}$$

令其为0, 得驻点 $\begin{cases} (2k\pi, 0) \\ ((2k+1)\pi, -2) \end{cases}$

$$A = z''_{xx} = -(1 + e^y) \cos x$$

$$C = z''_{yy} = e^y (\cos x - y - 2)$$

$$B = z''_{xy} = -e^y \sin x$$

对 $(2k\pi, 0)$

则 $B^2 - AC < 0$. 又 $A = -(1 + e^y) \cos x < 0$. 则为极大值点, 极大值为 2

对 $((2k+1)\pi, -2)$

则 $B^2 - AC > 0$ 非极值点

eg2: $x^2 + y^2 + 4z^2 = 2$ 被 $x + y - z = 0$ 截得椭圆.
求其面积

$$\begin{cases} x^2 + y^2 + 4z^2 = 2 \\ x + y - z = 0 \end{cases}$$

取 $P(x, y, z)$ 到原点的距离 $d^2 = x^2 + y^2 + z^2$.

则问题化为条件极值.

由 Lagrange:

$$\text{则可设 } F(x, y, z) = x^2 + y^2 + z^2 + \lambda(x^2 + y^2 + 4z^2 - 2) + \mu(x + y - z)$$

$$\text{令 } F'_x = F'_y = F'_z = F'_\lambda = F'_\mu = 0, \text{ 得条件极值点}$$

$$\Rightarrow \begin{cases} |x| = \frac{1}{3} \\ |y| = \frac{1}{3} \\ |z| = \frac{2}{3} \end{cases} \Rightarrow r^2 = \frac{2}{3} \quad \text{或} \quad \begin{cases} |x| = 1 \\ |y| = 1 \\ |z| = 0 \end{cases} \Rightarrow r^2 = 2$$

$$\text{则 } S = \pi ab = \pi \cdot \sqrt{\frac{4}{3}} = \frac{2\sqrt{3}}{3} \pi$$

5. 方向导数.

$$\frac{\partial u}{\partial l} \Big|_P = \frac{\partial u}{\partial x} \cos \alpha + \frac{\partial u}{\partial y} \cos \beta + \frac{\partial u}{\partial z} \cos \gamma = \text{常数}$$

其中 $(\cos \alpha, \cos \beta, \cos \gamma)$ 为 $\vec{l}$ 的单位向量.